

Day_1_Bell_state

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QUANTUM AI LABS DAY-1: BELL STATE VERIFICATION: RAJIV KUMAR YADAV

```
[19]: from qiskit import QuantumCircuit
      from qiskit_aer import AerSimulator
      import matplotlib.pyplot as plt
```

Simulator

```
[20]: simulator = AerSimulator()

shots = 1000
```

1 Bell State 1: $|\Phi^+\rangle = (|00\rangle + |11\rangle) / \sqrt{2}$

```
[21]: phi_plus = QuantumCircuit(2, 2)

phi_plus.h(0)
phi_plus.cx(0, 1)
phi_plus.measure([0, 1], [0, 1])
phi_plus.draw()
```

```
[21]:
q_0:  H      M

q_1:      X      M

c: 2/
      0  1
```

2 Bell State 2: $|\Phi^-\rangle = (|00\rangle - |11\rangle) / \sqrt{2}$

```
[22]: phi_minus = QuantumCircuit(2, 2)

phi_minus.h(0)
phi_minus.z(0)
phi_minus.cx(0, 1)
phi_minus.measure([0, 1], [0, 1])
```

```
phi_minus.draw()
```

[22]:

```
q_0:  H    Z    M

q_1:      X    M

c: 2/

      0  1
```

3 Bell State 3: $|\Psi^+\rangle = (|01\rangle + |10\rangle) / \sqrt{2}$

```
[23]: psi_plus = QuantumCircuit(2, 2)

psi_plus.h(0)
psi_plus.x(1)
psi_plus.cx(0, 1)
psi_plus.measure([0, 1], [0, 1])
psi_plus.draw()
```

[23]:

```
q_0:  H      M

q_1:  X  X    M

c: 2/

      0  1
```

4 Bell State 4: $|\Psi^-\rangle = (|01\rangle - |10\rangle) / \sqrt{2}$

```
[24]: psi_minus = QuantumCircuit(2, 2)

psi_minus.h(0)
psi_minus.x(1)
psi_minus.z(0)
psi_minus.cx(0, 1)
psi_minus.measure([0, 1], [0, 1])
psi_minus.draw()
```

[24]:

```
q_0:  H    Z    M

q_1:  X      X    M

c: 2/

      0  1
```

5 Store circuits

```
[25]: bell_states = {  
    "Phi+": phi_plus,  
    "Phi-": phi_minus,  
    "Psi+": psi_plus,  
    "Psi-": psi_minus  
}
```

6 Run all four Bell states

```
[26]: results = {}  
  
for name, circuit in bell_states.items():  
    job = simulator.run(circuit, shots=shots)  
    result = job.result()  
    counts = result.get_counts()  
  
    results[name] = counts  
  
    print(f"\n{name}")  
    print("-" * 20)  
    print(counts)
```

Phi+

{'00': 508, '11': 492}

Phi-

{'11': 512, '00': 488}

Psi+

{'10': 488, '01': 512}

Psi-

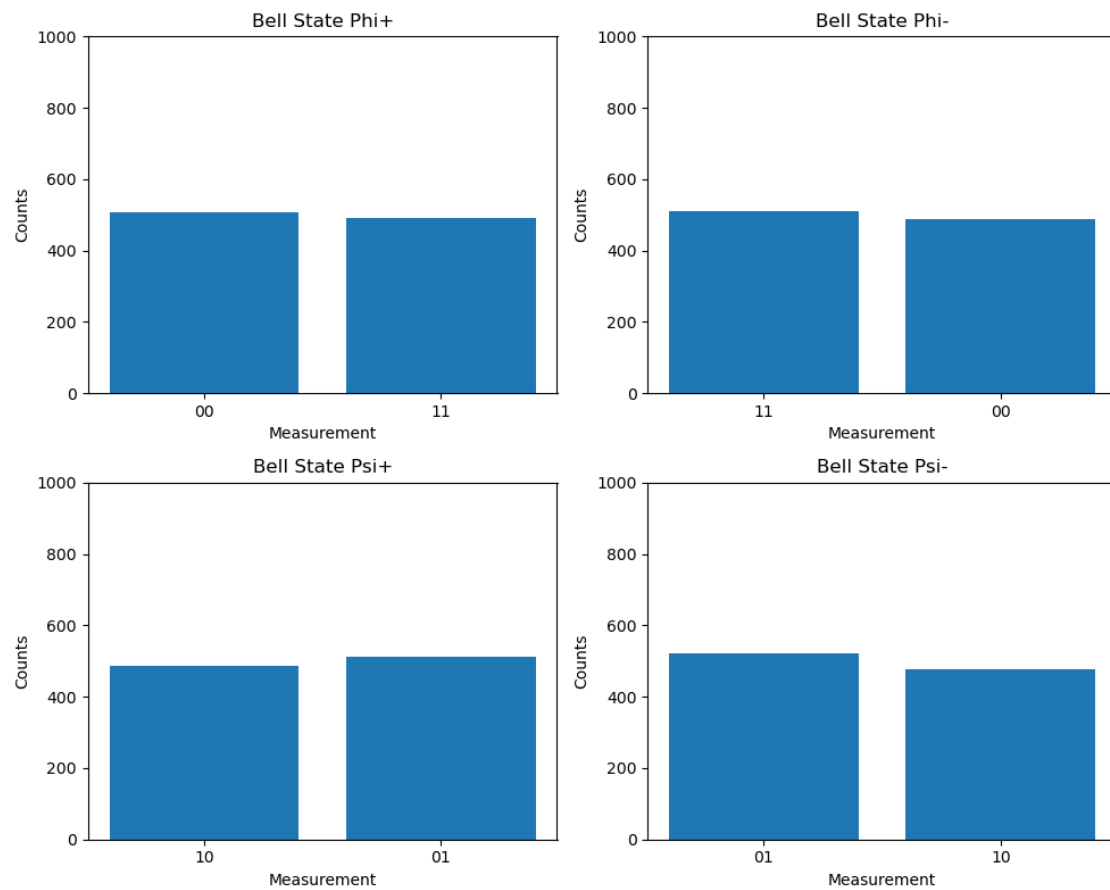
{'01': 523, '10': 477}

Plot measurement results

```
[27]: fig, axes = plt.subplots(2, 2, figsize=(10, 8))  
  
for ax, (name, counts) in zip(axes.flatten(), results.items()):  
    ax.bar(counts.keys(), counts.values())  
    ax.set_title(f"Bell State {name}")
```

```
ax.set_xlabel("Measurement")
ax.set_ylabel("Counts")
ax.set_ylim(0, shots)
```

```
plt.tight_layout()
plt.show()
```



[]: